CLAIMS

What is claimed is:

1. A method for transmitting data over a parallel communication interface having a plurality of physical channels, each capable of representing N unique states, the method comprising the steps of:

    - providing a pre-defined legacy character set containing a plurality of characters;
    - creating an encoding map that maps each character of the legacy character set to a unique state of the N states;
    - receiving an input data stream comprising a sequence of characters from the legacy character set; and
    - for each character in the sequence, transmitting it by setting a single physical channel to the corresponding unique state defined in the encoding map.

2. The method of claim 1, wherein the legacy character set is a Unicode standard, such as UTF-8 or UTF-16.

3. The method of claim 1, wherein the physical channels are pixels and the N unique states are pixel colors.

4. The method of claim 1, wherein a plurality of characters are transmitted simultaneously in a single frame by using a plurality of physical channels.

5. The method of claim 1, wherein the parallel communication interface is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.